

TGI: Topological Geometric Intelligence

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Abstract

Topological Geometric Intelligence (TGI) is proposed as an independent computational research program in which raw character identity, source-owned experience, discrete spatial organization, and certified traversal form a continuous learning and resolution chain. Its starting point is the NINMENI axiom that one Unicode character has one static identity. TGI derives its own objects and operators from the goal of forming inspectable, revisable knowledge rather than transplanting an existing neural implementation. This paper states the paradigm contract and connects five separately specified mechanisms: character identity and crystal-lattice addressing; C/L-band-owned phase and organization; interventional formation of spatial referents; deterministic resolution with continual revision; and owned spatial execution. It distinguishes a stored route from a source-valid organization and an intervention-supported relation. The published experimental set includes exact two-stream reconstruction, phase and counterexample controls, interventional physical traces, complete-frontier resolution checks, checkpoint recovery, and locked scale measurements. Each result is tied to a specific contract and falsifier. This foundation paper defines how the components jointly constitute TGI; each technical paper supplies its own formal details and reproducibility path.

1. Research question and position

The central question is whether a machine can form, revise, and use knowledge through discrete identity and geometry while preserving an inspectable chain from experience to answer. A successful answer must identify the source of a claim, show how its organization formed, expose the route by which a query reaches it, and state when no complete route is available. TGI treats this as an architectural question, not a renaming of an existing model.

The original *KONSEP.txt* places TGI alongside machine learning as a proposed AI paradigm, with derived subfields beneath it. That is a research taxonomy: its legitimacy comes from independently specified mechanisms and their falsifiers. Earlier six-branch labels in the manifesto describe exploration, not a requirement that six software modules or six papers must exist. The present publication series instead follows mechanism ownership, as the four NINMENI reference preprints do. This foundational paper owns the joint contract; its companions each own a narrower technical question.

TGI adopts the NINMENI identity axiom without copying a VEYRA NYRA model. Within the TGI foundation, subword segmentation, gradient descent, softmax scaled dot-product attention, and continuous vector embeddings are excluded as learning or resolution operators. Programming languages, arrays, indexes, hashes used with equality checks, and compiled kernels are neutral implementation instruments. They become relevant to the paradigm only through the identity, ownership, and provenance invariants they preserve.

2. The common objects and their invariants

Let a raw stream be an ordered sequence of Unicode scalar values. The TGI codec assigns one static ID to each scalar, currently $ord(c)+16$, with 0-15 reserved. Distinct occurrences of the same character retain that identity while acquiring distinct source, time, and position records. No phrase ID or subword ID replaces the constituent identities. Event order is part of the observable input and must survive checkpoint replay.

An admitted frame consists of these occurrences, source provenance, and a bounded event interval. A crystal address is discrete and is allocated to an occurrence in a context lane; an adjacency bond can connect only

source-valid neighbors. The five-coordinate layout in the present implementation is a specific addressing contract, not a proof that every conceptual relation is intrinsically five-dimensional. Complete traversal requires an explicit end condition. Missing, ambiguous, and incomplete routes are distinguishable outcomes, because a work ceiling cannot certify absence.

The C bands own local adjacency positions and the L bands own temporal coverage. An occurrence phase receipt, denoted Ω and ξ in the technical paper, checks structural admission. Organization-level Ω_H instead counts distinct realizations across source-valid episodes. These values have different types and purposes; a large scalar by itself is not knowledge. A counterexample can revoke an organization without deleting the historical events from which it formed. Source validation recomputes ownership and phase receipts before a stored claim is allowed to authorize an answer.

Knowledge in this architecture is consequently not just a lattice edge. The joint record is a source-valid organization with an explicit domain of application, support, counterexamples, phase state, and a verifiable route to an output or action. Physical referent relations add an intervention record: the system changes an actuator variable, observes the resulting sensor trace, and compares it with the unaltered course. A repeated linguistic surface alone does not establish that two descriptions share a world referent.

3. Derivation of the TGI chain

The derivation starts with the goal of preserving exact experience. Static character identity and ordered occurrences give the substrate. Source-owned C/L organization then determines which local and long-range relations were actually witnessed. Discrete address allocation makes those witnesses retrievable without replacing the characters with continuous learned tokens. Phase admission prevents an unverified receipt from becoming a crystal.

The next goal is useful knowledge beyond verbatim replay. Organization is formed from contrasted episodes, including what remains fixed and what varies. It is eligible only under its stated coverage and evidence rule. The physical branch tests a further step: whether changes in the world support a relation that survives a novel intervention. Observed counterexamples can fracture a class and trigger a revised organization; they are retained as evidence, not averaged away by a score.

Finally, a query must resolve through a complete, inspectable path. The resolution engine evaluates admissible alternatives and source dependencies, including guarded transformations and actual feedback from action. A certificate records why a selected output is authorized. A rejection certificate must account for the full relevant frontier; an exhausted budget produces INCOMPLETE rather than a false claim that no route exists. Atomic checkpoint and recovery preserve this decision chain over time. Spatial execution work studies the cost of these same owned computations without changing their semantics.

4. Technical division of the publication series

Independent technical paper	Mechanism it owns	Main falsifier
Character Identity and Discrete Crystal-Lattice Addressing	NMU occurrences, five-coordinate lane, exact route	repeated prefix crosses lanes or replay changes allocation
Band-Owned Phase Formation and Source-Valid Organization	C/L ownership, phase receipt, raw organization	invalid receipt enters the active crystal
Interventional Formation and Fracture of Spatial Referent Classes	edit contrast, sensor response, class fracture	held-out intervention violates the formed relation
Deterministic Resolution and Continual Learning	guarded closure, complete frontier, feedback and recovery	uncertified selection or false NO_PATH after incomplete search
Owned Spatial Execution	semantic parity and cost of state-owned kernels	speed claim lacks locked protocol or changes output

The papers are not an ordinal maturity ladder. An identity or phase mechanism is not itself a theory of referents; an execution kernel is not an independent decision rule. Each paper contains definitions, method, result, limitation, and reproduction instructions for its own operator. The current series and its hash index are in *paper/technical/README.md* and *paper/technical/validation.json*.

5. Method, evidence, and falsification

The acceptance seed is deliberately small and adversarial. Two raw streams, "SISTEM TGI AMAN DARI HALUSINASI" and "SISTEM REAKTOR NUKLIR MEMBUTUHKAN PENDINGIN", share a prefix but must remain in separate crystal lanes. A query anchored to the first or second lane must reproduce its entire stream character for character; an unformed third anchor must return no path. This tests address and traversal isolation. It does not substitute for the separate test of knowledge formation.

The phase and organization work tests source-valid receipts, positive support, counterexamples, and recovery of the same state after reload. Interventional work adds signed physical captures, edit transport to unseen prefixes, class-fracture controls, and independent checking of source and action provenance. Resolution work tests every relevant alternative, distinction of NO_PATH from INCOMPLETE, dependency validity, actual feedback, and atomic recovery. The execution study fixes input sizes and compares a state-owned implementation with its prior version under parity checks. These are different evidential claims and cannot be exchanged for a single accuracy number.

The historical foundation manuscript *paper/TGI_FOUNDATION_DRAFT.md* records the longer development chain and negative results. The compact publication set references exact contracts and receipts rather than reproducing a large test log in prose. The complete evidence bundle is indexed in *publication/TGI_V1_20261004/RELEASE.json*; its archive members and hashes are independently verifiable. Local validation of the six manuscript/PDF pairs and the publication ZIP is recorded in the series release metadata.

6. Scope of the resulting claim

TGI's defining claim is mechanistic: raw identity and experience ownership can support a connected cycle of formation, intervention, revision, and certified resolution without making a probabilistic next-token choice the governing operator. The component papers give the concrete laws and tests of that cycle. Exact path reconstruction, source validity, physical action traces, and scale parity are each stated at their own experimental boundary. No scalar phase metric alone proves truth in the world, and a closed route claim applies to the complete frontier certified for a query. This explicit division makes the architecture open to decisive counterexamples and permits each mechanism to improve without silently changing the common axiom.

References

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